

# AST 6007-01 Spacecraft Dynamics and Control

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## Final Examination

December 15, 2014 1:00-2:50 PM

Close Books/Close Notes

In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

$$\omega_0 = \frac{2\pi}{24}$$

1. (40) Consider a satellite initially on a geosynchronous circular equatorial orbit whose constant orbital frequency is denoted as  $\vec{\omega} = \omega \vec{k}$ . After performing a maneuver, it is now located at the following relative position and velocity with respect to its non-maneuvering position in the local-vertical-local-horizontal (LVLH)  $\vec{i}\vec{j}\vec{k}$ -frame attached to the geosynchronous orbit:

$$(x, y, z) = (-110, 70, 10) \text{ km}$$

$$(\dot{x}, \dot{y}, \dot{z}) = (-10, 20, 1) \text{ m/sec}$$

If the satellite is to initiate a two-impulse maneuver to return to its original non-maneuvering location in 2 hours, what are the changes in velocities at the initial and terminal time, i.e.,  $\Delta \vec{v}_0$  and  $\Delta \vec{v}_f$ ?

Hint: The solution to the Hill-Clohessy-Wiltshire (HCW) equation is given as

$$x(t) = \frac{\dot{x}_0}{\omega} \sin(\omega t) - \left( 3x_0 + \frac{2\dot{y}_0}{\omega} \right) \cos(\omega t) + \left( 4x_0 + \frac{2\dot{y}_0}{\omega} \right)$$

$$y(t) = \left( 6x_0 + \frac{4\dot{y}_0}{\omega} \right) \sin(\omega t) + \frac{2\dot{x}_0}{\omega} \cos(\omega t) - (6\omega x_0 + 3\dot{y}_0)t + \left( y_0 - \frac{2\dot{x}_0}{\omega} \right)$$

$$z(t) = \frac{\dot{z}_0}{\omega} \sin(\omega t) + z_0 \cos(\omega t)$$

If the relative position in the HCW equation is equal to zero after time  $t$ , the initial position and velocity must satisfy the following relation:

$$\dot{y}_0 = \frac{[6x_0(\omega t - \sin(\omega t)) - y_0]\omega \sin(\omega t) - 2\omega x_0(4 - 3\cos(\omega t))(1 - \cos(\omega t))}{(4\sin(\omega t) - 3\omega t)\sin(\omega t) + 4(1 - \cos(\omega t))^2}$$

$$\dot{x}_0 = -\frac{\omega x_0(4 - 3\cos(\omega t)) + 2(1 - \cos(\omega t))\dot{y}_0}{\sin(\omega t)}$$

$$\dot{z}_0 = -z_0 \omega \cot(\omega t)$$

2. (20) Consider the Euler's rotational equations of motion represented in the principal axes of body-fixed frame:

$$\begin{cases} I_x \dot{\omega}_x = (I_y - I_z)\omega_y \omega_z + M_x \\ I_y \dot{\omega}_y = (I_z - I_x)\omega_x \omega_z + M_y \\ I_z \dot{\omega}_z = (I_x - I_y)\omega_x \omega_y + M_z \end{cases}$$

- A. (6) For a general asymmetric spacecraft, show that  $(\omega_x \equiv 0, \omega_y \equiv 0, \omega_z \equiv \bar{\omega}_z = \text{constant})$  is an equilibrium solution with no external moments.
- B. (8) Linearize the Euler's rotational equations of motion with respect to the equilibrium solution given in (a).
- C. (8) Derive the characteristic equation for two non-trivial equations obtained in (b) and their associated eigenvalues case-by-case (i.e., depending on the magnitude of moment of inertia component).

3. (20) Consider the pitch axis control model for a spacecraft in a circular orbit given by

$$I_y \Delta \ddot{\theta} = 3\omega_0^2 (-I_x + I_z) \Delta \theta + M_y$$

where the orbital frequency  $\omega_0 = 0.0012 \text{ rad/sec}$ , the spacecraft principal moments of inertia are  $I = \text{diag}(150, 100, 200) \text{ kg-m}^2$ . In this equation  $\Delta \theta$  denotes the pitch angle and  $M_y$  denotes the pitch axis control moment.

- A. If the pitch axis control moment  $M_y = 0$ , show that the pitch equilibrium  $(\Delta \theta, \Delta \dot{\theta}) = (0, 0)$  is an unstable equilibrium.
- B. Show that linear state feedback to the pitch axis control moment can be used to stabilize the pitch axis equilibrium  $(\Delta \theta, \Delta \dot{\theta}) = (0, 0)$  of the spacecraft. Choose feedback gains so that the equilibrium of the closed loop is asymptotically stable with damping ratio of 0.7 and natural frequency of  $0.008 \text{ rad/sec}$  (that is, the closed loop characteristic polynomial is  $s^2 + 0.0112s + 0.000064$ ).

4. (20) A quaternion vector  $(q_1, q_2, q_3, q_4)$  satisfies the following rotational kinematic equations

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \frac{1}{2} S(\omega) \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} + \frac{1}{2} q_4 \omega$$

$$\dot{q}_4 = -\frac{1}{2} \omega^T \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}$$

Show that if the initial values of the quaternion satisfy

$$q_1^2(0) + q_2^2(0) + q_3^2(0) + q_4^2(0) = 1$$

then

$$q_1^2(t) + q_2^2(t) + q_3^2(t) + q_4^2(t) = 1$$

holds for all  $t \geq 0$ .